ii) Common-Emitter (CE) configuration

In this configuration, the emitter is common to both the input and output circuits as shown in Figure 10.27. The base current I_B is the input current and the collector current I_C is the output current. The input signal is applied between emitter and base while the output is measured between collector and emitter.

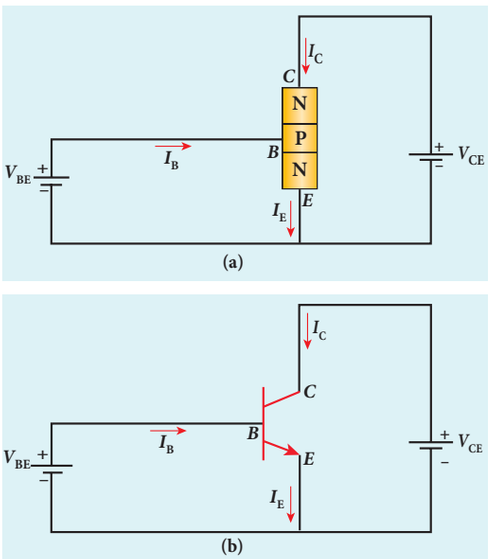


Figure 10.27 NPN transistor in common emitter configuration (a) Schematic circuit diagram (b) Circuit symbol

iii) Common-Collector (CC) configuration

Here, the collector is common to both the input and output circuits as shown in Figure 10.28. The base current I_B is the input current and the emitter current I_E is the output current. The input signal is applied between base and collector while the output is measured between emitter and collector.

Note As the output is taken from the emitter in common collector configuration, it is called an emitter follower.

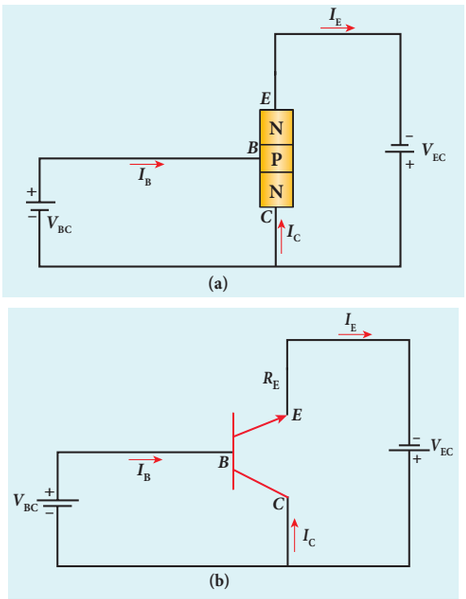


Figure 10.28 NPN transistor in common collector configuration
(a) Schematic circuit diagram
(b) Circuit symbol

10.4.2 Transistor action in the common base mode

The operation of an NPN transistor in the common base mode is explained below. The current flow in a common base NPN transistor in the forward active mode is shown in Figure 10.29.

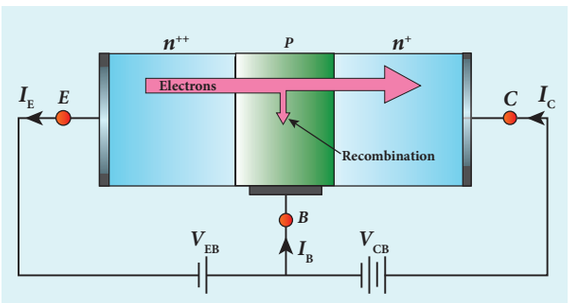


Figure 10.29 Flow of current in a NPN transistor

Basically, a BJT can be considered as two *p-n* junction diodes connected back-to-back. In the forward active bias of the

UNIT-10_XII-Physics_Vol-2_New.indd 217 21-07-2020 15:35:24